

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

Nov 2012

EE208 Micro Electronics & Integrated Circuit

Date: 30.11.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
- 5.

SECTION – I

Q - 1 What are the electrical characteristics of an ideal op-amp? Explain equivalent circuit of op-amp with ideal voltage transfer curve. [07]

Q - 2 (a) Draw pin diagram and block diagram of 555 timer IC. Also Give description of each pin. [10]

(b) Explain pin configuration of 741 op-amp IC. [04]

OR

Q - 2 (a) Explain Astable operation of 555 IC. [05]

(b) Explain operating principle of Phase-locked loop with block diagram. [04]

(c) Derive equation of closed-loop voltage gain (A_F) for Voltage-series feedback amplifier with circuit diagram. [05]

Q - 3 (a) Explain Summing, Scaling and Averaging Amplifier in inverting configuration. [07]

(b) Explain first order low-pass Butterworth filter. [07]

OR

Q - 3 (a) Define the terms: (i) Input offset voltage (ii) Input offset current (iii) CMRR (iv) SVRR (v) Output voltage swing [05]

(b) Explain differential amplifier with one op-amp. [04]

(c) Explain all-pass filter with neat diagram. [05]

SECTION - II

Q - 4 What is a Transistor? Draw and explain characteristics of common base connection of a transistor. [07]

Q - 5 (a) Compare Different connections of a Transistor. [06]

(b) Find the (i) input impedance and (ii) voltage gain for the circuit shown in Figure-1. [08]
(Use only h parameter equations)

OR

Q - 5 (a) What are the basic conditions for achieving faithful amplification of a transistor? Explain each. [07]

(b) Figure-2 shows biasing with base resistor method. (i) Determine the collector current I_C and collector-emitter voltage V_{CE} . Neglect small base-emitter voltage. Given that $\beta=50$. (ii) If R_B in this circuit is changed to $50\text{ K}\Omega$, find the new operating point. [07]

Q - 6 (a) What do you mean by h parameter? Draw its equivalent circuit. Find out equations for input impedance, voltage gain and current gain of a transistor in terms of h parameter. [07]

(b) Define Shorted-gate drain current, Pinch off voltage and Gate-source cut off voltage for a JFET. [07]

OR

Q - 6 (a) What is JFET? Draw and explain its construction details. Also draw schematic symbol of JFET. [07]

(b) Draw and explain JFET biasing. [07]

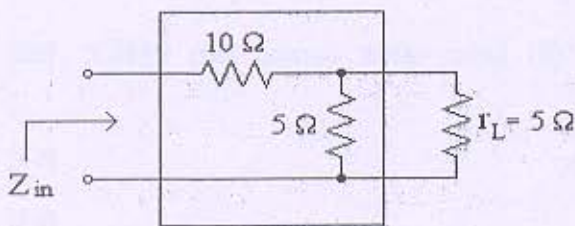


Figure-1

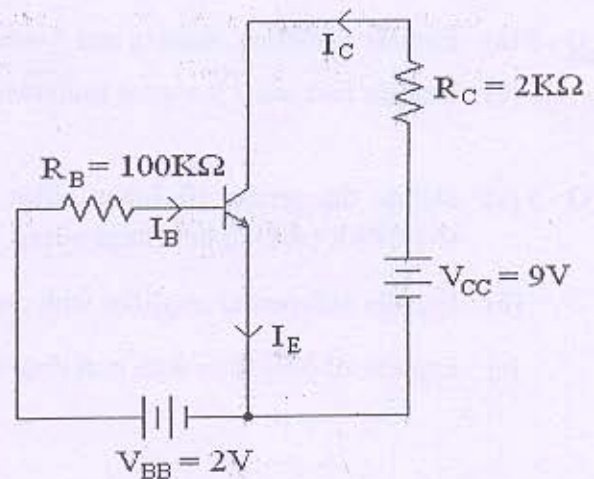


Figure-2
